External Validation and PRESS

Applied Linear Models I • Lecture 13 of 16

Applied Linear Models I

Today

- **PRESS**: leave-one-out prediction error in one fit.
- The leverage shortcut = brute-force LOO.
- **Predicted R^2** vs ordinary R^2.
- Train/test validation and the train–test gap.
- **Over-fit models validate worse.**

PRESS and the leverage shortcut

$$\text{PRESS} = \sum_{i=1}^{n} \left(y_i - \hat{y}_{(i)}\right)^2, \qquad y_i - \hat{y}_{(i)} = \frac{e_i}{1 - h_{ii}}.$$

$$\boxed{\text{PRESS} = \sum_{i=1}^{n} \left(\frac{e_i}{1 - h_{ii}}\right)^2}$$

- **One fit**, not n – just residuals and leverages.
- High-leverage points ($h_{ii} \to 1$) dominate PRESS – links to influence (L7).
- Worked check: shortcut = brute-force LOO exactly (245.483).

Predicted R^2

$$R^2_{\text{pred}} = 1 - \frac{\text{PRESS}}{\text{SST}} \leq R^2.$$

`mpg ~ wt + hp + qsec`

$R^2 = 0.835$ vs $R^2_{\text{pred}} = 0.782$ – a modest gap (mild optimism).

A **large** gap warns of over-fitting. R^2_{pred} cannot be gamed by adding predictors – unlike R^2.

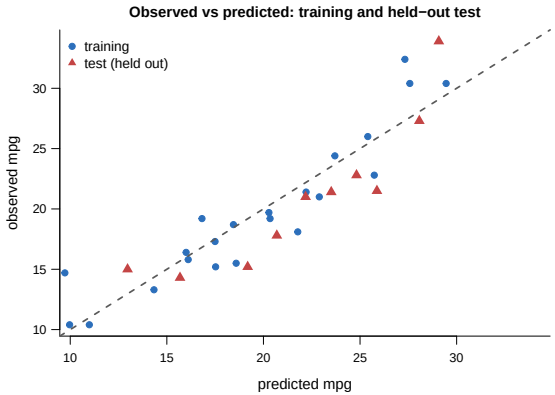
Train/test validation

Fit on training, evaluate on held-out test:

$$R^2_{\text{valid}} = 1 - \frac{\text{SSE}_{\text{test}}}{\text{SST}_{\text{test}}}.$$

`mtcars` 70/30:

- training $R^2 = 0.858$
- validation $R^2 = 0.751$
- gap ≈ 0.11 (normal)

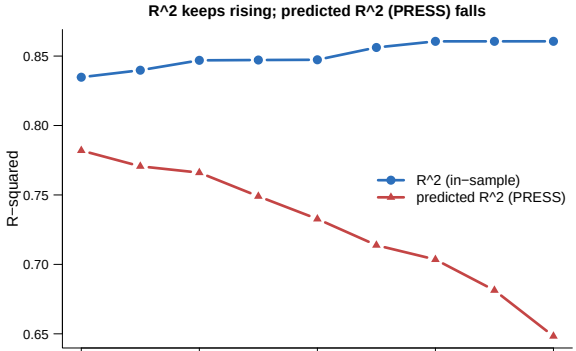


Keep the test set untouched during modeling.

Over-fit models validate worse

Add six **pure-noise** predictors:

Model	R^2	PRESS
lean (wt+hp+qsec)	0.835	245.5
bloated (+6 noise)	0.861	333.9



Summary

- PRESS $= \sum (e_i/(1-h_{ii}))^2$ – LOO prediction error, one fit.
- Predicted $R^2 = 1 - \text{PRESS}/\text{SST} \leq R^2$; large gap = over-fit.
- Train/test: held-out R^2; small gap normal, large gap not.
- Adding noise raises R^2 but worsens PRESS.

Next

Lecture 14: weighted and generalized least squares (non-spherical errors).